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"on my honor, I have neither given nor reciarved any unauthorized and/or inapperopriate assistance for all sessions of that exam. the work done on that exam is totally my own. I understand that by the school code. violation of these prenciples will lead to a zero grade and is subject to harsh discipline isunsi?

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1).- $$A = \begin{bmatrix} 1 & \alpha & -Y & \alpha \\ 3\alpha & 5 & -9Y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & K & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{bmatrix}$$

$$\frac{2R}{3} \Rightarrow A \begin{bmatrix} 1 & \alpha & -y & \alpha \\ \cancel{\alpha} & \frac{5}{3} & -3Y & \frac{1}{3} \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & -2 \end{bmatrix}$$

reduced after that making equations

with all veriables.

after that we can find the values.

3). $\boxed{a)}$ $H = I - 2uu^T$

$H^2 = (I - 2uu^T)^2$

$= I^2 - 4uu^T + 4uu^T uu^T$

u is unit vector $u^Tu = 1$

$H^2 = -4uu^T + 4u(1)u^T$

$= I^2 - 4uu^T + 4uu^T$

$= I^2 = I$

$\boxed{b)}$ $H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2uu^T$

$= H$

$H^2 = HH^T$

$\Rightarrow HH^T = I$ $\Rightarrow$ H is orthonormal matrix.

$\boxed{c)}$ $Hu = (I - 2uu^T)u$

$= u - 2u(u^Tu)$

$= u - 2u$

$= -u = (-1)u \Rightarrow$ $\boxed{\text{eign value is } -1}$

$\boxed{D)}$ H is diagonallable ~~v is~~ v is orthognal to u

$\Rightarrow u^Tv = 0$ $Hv = (I - 2uu^T)v = v - 2u(u^Tv)$
$= v$

so eisen value associated v is 1
the orthognal complement of u has dimension $(n-1)$
there are n independent eigenvectors of H, (H is diagonallable

4)- a)- The equation that is setting the processing is:

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

b)-